

About the data: threatened species table

Background for [Homework 1](#). Read this before you start the tasks.

The table

The file is a gzip-compressed, comma-separated table with one species per row and a header line. The columns are:

#	column	#	column
1	taxonid	8	scientific_name
2	kingdom_name	9	taxonomic_authority
3	phylum_name	10	infra_rank
4	class_name	11	infra_name
5	order_name	12	population
6	family_name	13	category (IUCN status, e.g. CR, EN, VU)
7	genus_name	14	main_common_name

Look at it first, before writing any code (`zcat file | head, zless`). Note that the file is compressed: use `zcat` (or `gzip -dc`) to read it, and never decompress it into your repository (it is ~18 MB uncompressed, and you should not commit data files).

A warning about commas in the data

The file is comma-separated, but a few text fields themselves contain commas. When that happens the field is wrapped in double quotes so the comma is not a separator, for example (an illustrative row; the authority field is quoted):

```
...,Eugenia,Eugenia oreophila,"Smith, 2010",,,,LC,
```

`cut -d,` does not know about quotes: it splits "Smith, 2010" into two fields, and every column after it moves one to the right. Only 14 of the columns are *supposed* to exist, but about 80,000 of the rows contain a quote, so this matters. Columns 1-8 come before any quoted field and are always safe. Columns 9-14 (`taxonomic_authority`, `population`, `main_common_name`) can contain quoted text, so `cut -f13` for `category` will give wrong answers.

The simple fix: delete every quoted piece of text *before* you cut. The text inside quotes is never one of the fields you want to count here, and removing it leaves exactly 14 columns on every row.

remove "..." (anything between a pair of double quotes), then split on commas as usual

```
zcat threatened-species.csv.gz | tail -n +2 | sed 's/"[^"]*"//g' | cut -d, -f13 | sort | uniq -c | sort -
```

Use `sed 's/"[^"]*"//g'` any time you need a column after column 8. (The only column this fix cannot be used to *read* is `main_common_name`, since its quoted values get deleted too.) Check it worked: after the `sed`, `awk -F, '{print NF}' | sort | uniq -c` should print a single line saying 14.